

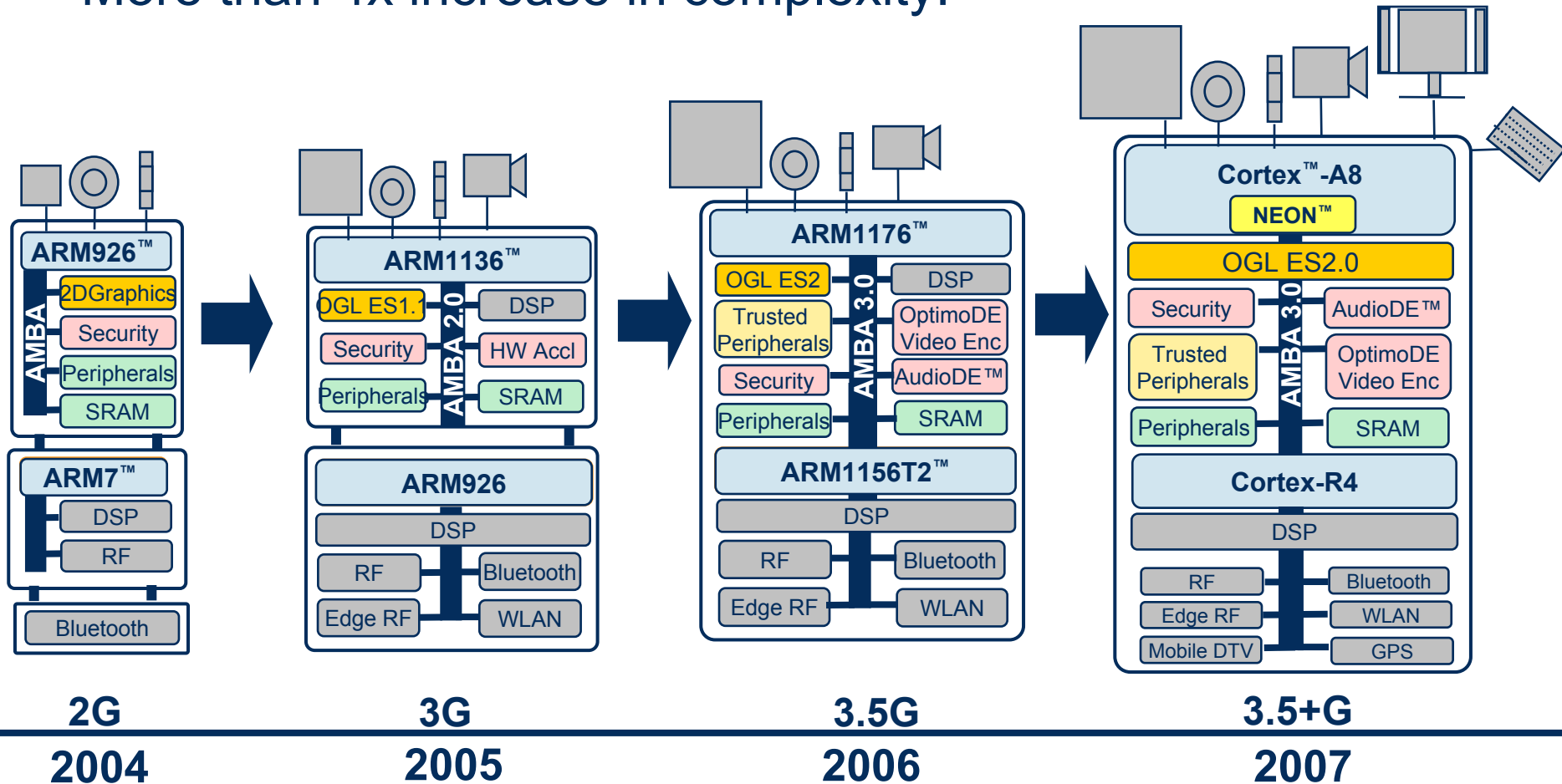
Anatomy of a High Performance Device

Kerry McGuire

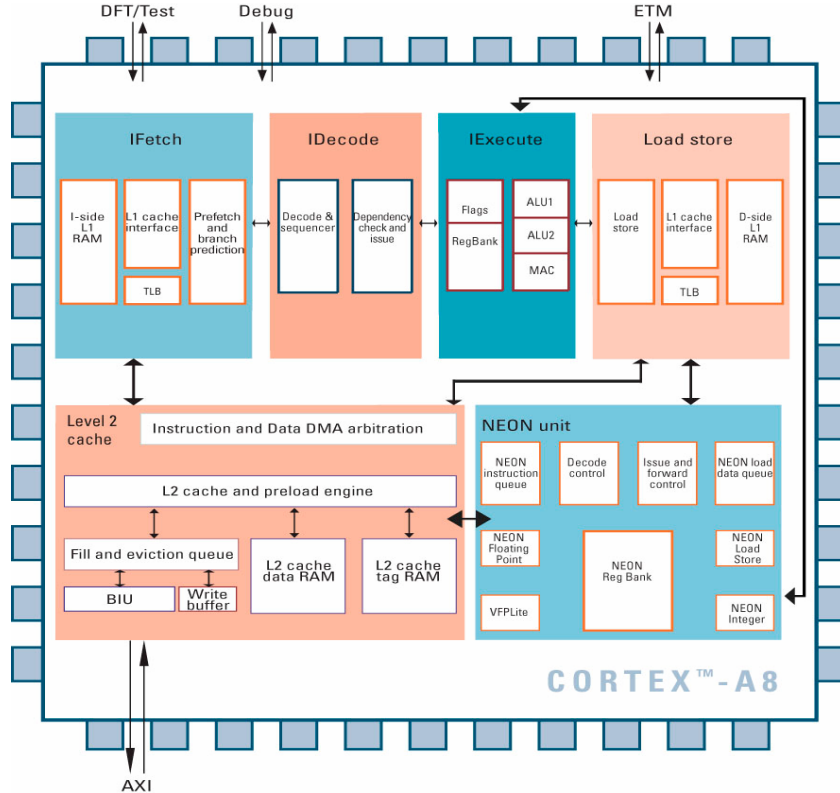
Product Manager, Cortex-A8 Processor
Processor Division

Advancing Mobile Platform Design

- Power, performance, and price have driven mobile platforms to industry leading levels of integration
- More than 4x increase in complexity!



Cortex-A8 Processor

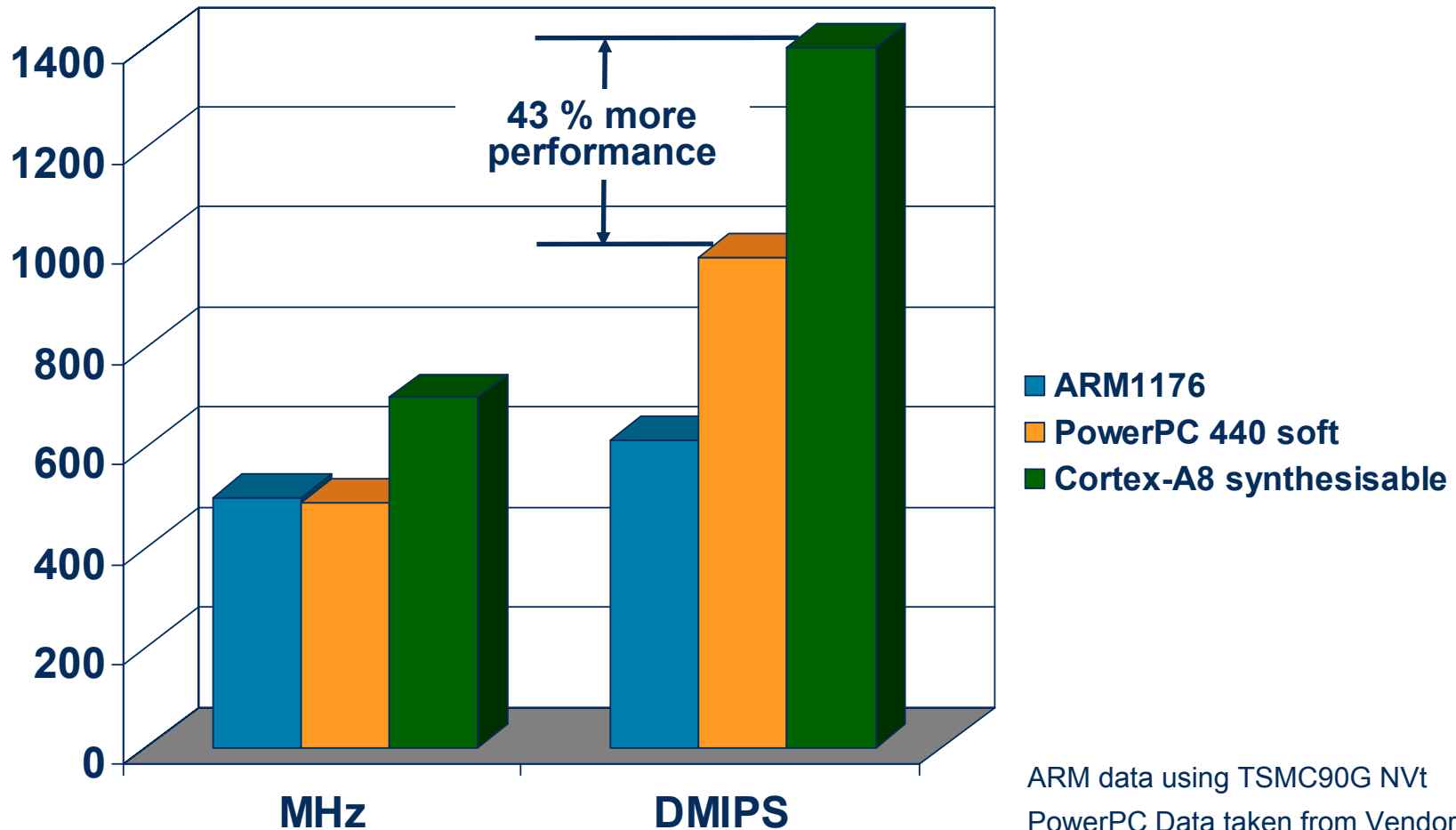


- ARMv7 architecture
 - NEON™ media acceleration
 - Thumb®-2 technology
- Advanced microarchitecture
 - 13-stage superscalar pipeline
 - Advanced branch prediction
- Optimised implementation
 - 1+ GHz and 2,000+ DMIPS on high performance technologies
 - 600+ MHz and < 300 mW on low leakage 65LP technology

- Introducing new synthesizable implementation flow
 - Flexible deployment and portability using advanced EDA tool flow
 - Synthesizable flow developed in collaboration with Cadence

Cortex-A8 Synthesisable Processor

- Faster time-to-market with competitive advantage



ARM data using TSMC90G NVt
PowerPC Data taken from Vendor Websites,
TSMC13LV data scaled to TSMC90G

AMBA®: Enabling Integration

- High performance, highly integrated systems require efficient on-chip fabric and interconnect technologies
- AMBA® 3 AXI™ enables efficient integration for next generation high performance, highly integrated systems

AMBA³

System Architecture

PrimeCell® peripherals

- Powerful highly configurable IP

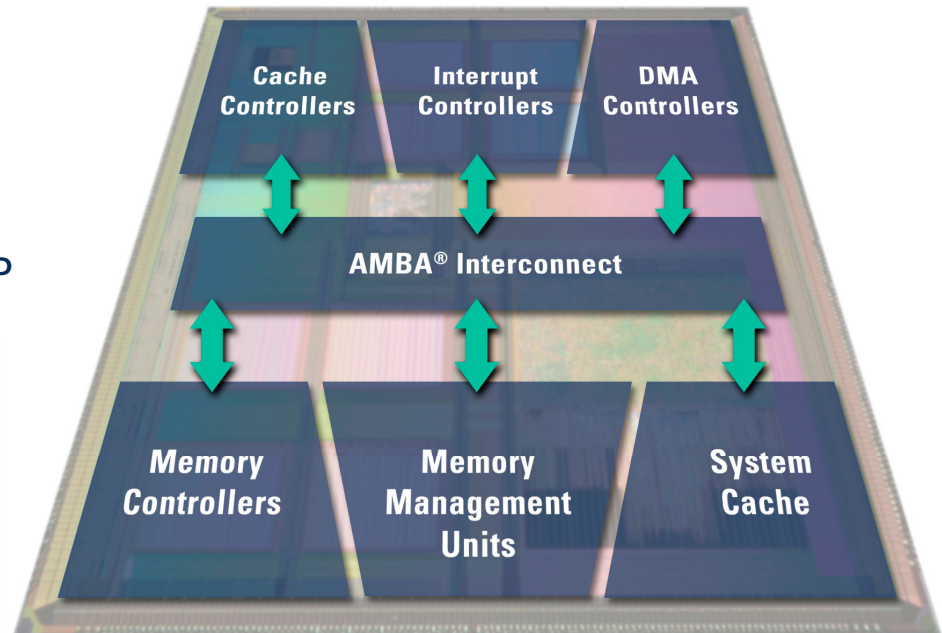
AMBA Designer

- Unique Ecosystem tool
- Design, Simulation, Analysis
- Delivers optimized IP (RTL)

AMBA specification

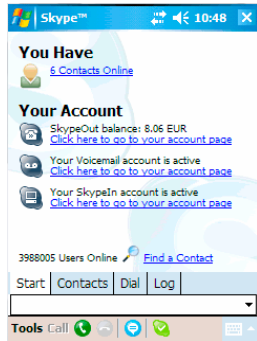
- Development and support of the AMBA SoC bus specification: “..... the *de facto* standard is of course the ARM bus architecture, AMBA.”

Ron Wilson, EETimes, 10/10/2005

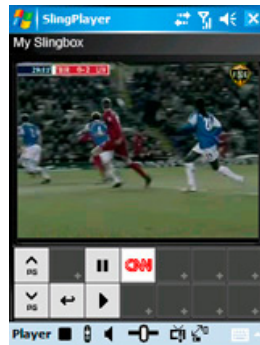


NEON: Ready for the Unexpected

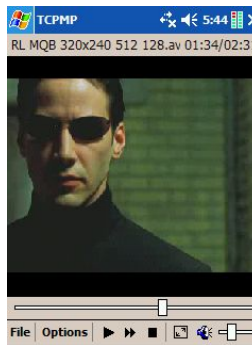
- Universal, easily accessible media acceleration for emerging applications and evolving standards in the mobile market



VoIP
Communications



Portable media player
for streaming from
home DVR

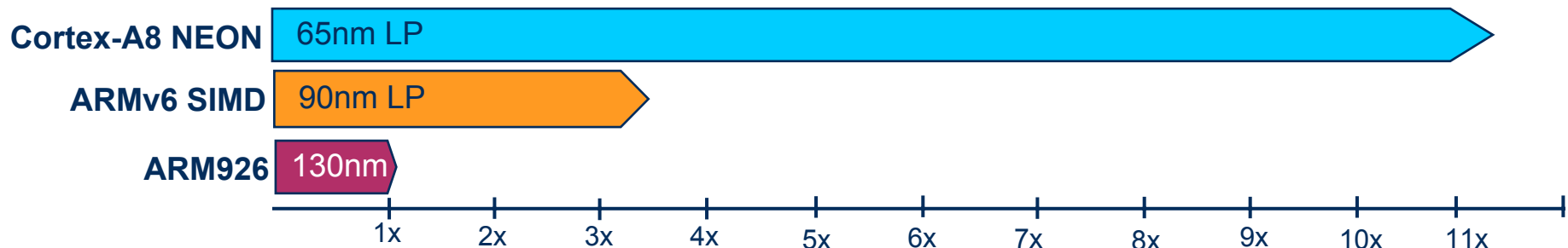


Free video player supporting
formats not included with
bundled video players.



Voice encryption for
mobile handsets.

More than 10x more media processing capabilities over 2 generations



Graphics Processor Integration

- Apps processors work in concert with graphics processors
- Off-loading graphics onto hardware enables power reduction
- Standard API's like D3D Mobile and OpenGL ES ease integration

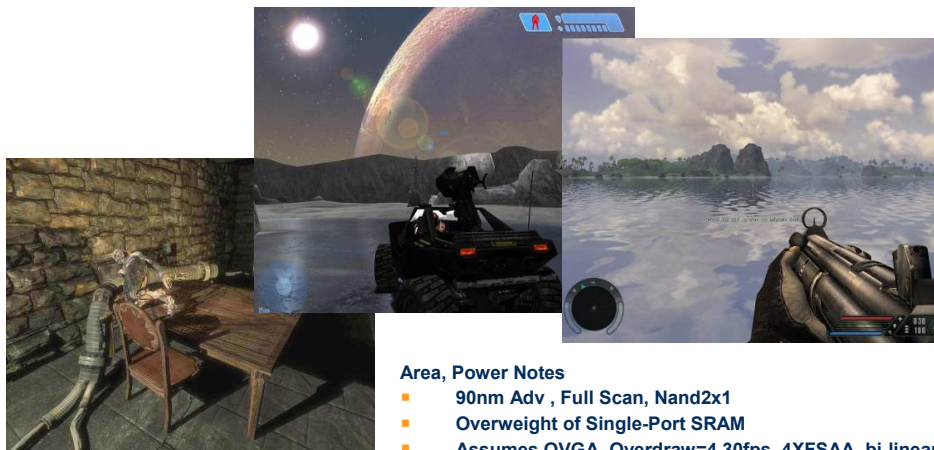
Mali 2xx Graphics Processors

✓ High quality results

Anti-aliasing, perspective correct textures, bilinear filtering, and bump mapping all costly/impractical if software-only solution

✓ Rich feature set for differentiation

Third generation IP cores takes 3D graphics beyond Shader Model 3.0



3rd Generation Cores

	Mali200	MaliGP2
Anti-Aliasing	4X / 16X	4X / 16X
OpenGL®ES 1.x	YES	YES
OpenGL®ES 2.x	YES	YES
SRAM	35kB	13kB
Die Area	5.5mm ²	1.5mm ²
Power	0.7mW/MHz	0.3mW/MHz
Max CLK	250MHz	250MHz
Fill rate / s	250 Mpix	NA
Triangles / s	8M	8M
DirectX9 Vista ext.	YES	YES

Rendering and Computation Equivalence Performance Notes

- Performance figures based on Max CLK
- 4X FSAA turned on (1 pix = 4 sub-samples) and bi-linear filtering
- Fill rate and geometry expressed as max theoretical performance

Take Full Advantage of Your Hardware

- ARM software optimised to take full advantage of the hardware capabilities while enabling a quick time-to-market



Fast, consistent execution
for Java applications



Secure, open
framework for
security applications



Execution of 3D content for
an outstanding gaming
experience



Run-time control for
maximising battery life

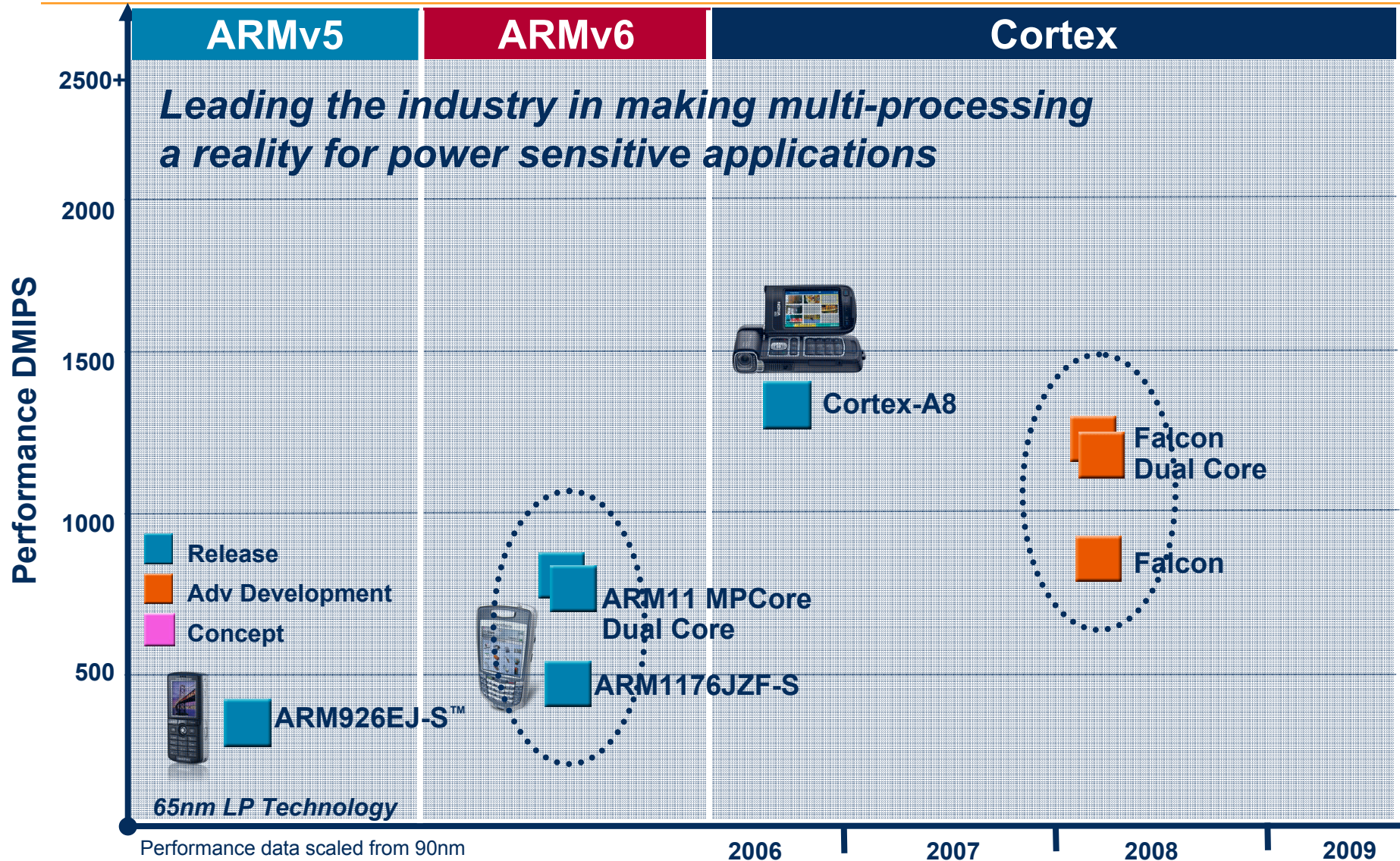
Enabling rich user experience to consumers

Next-Generation Mobile

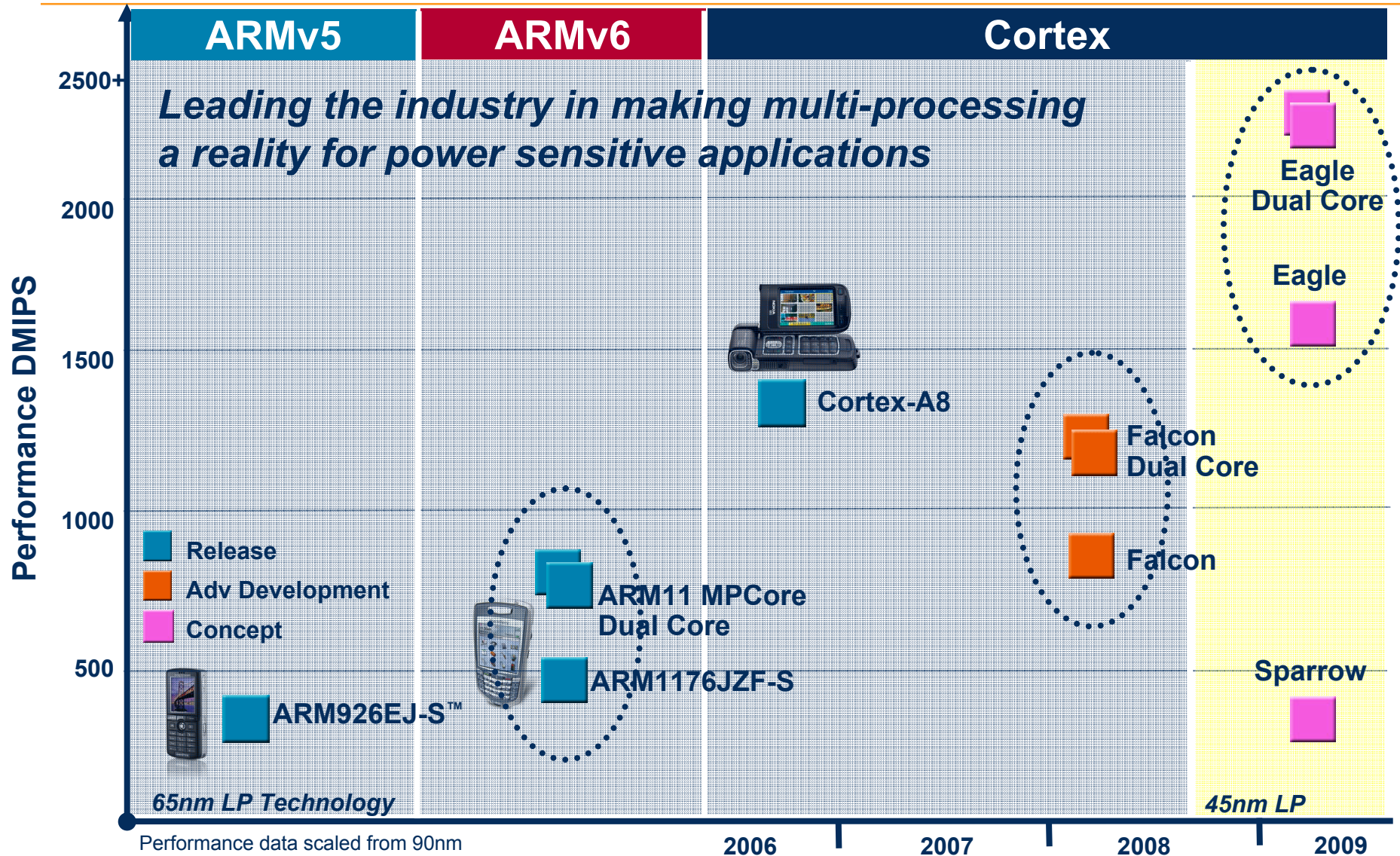
- Improve power consumption across all classes of processing
 - 2G/3G voice, data synchronization, MP3, general housekeeping
 - Data streaming, video, gaming, beyond 3G
- Balance performance improvements with area and power
 - Improved system-wide processing across multiple cores (including DSPs)
 - Improved execution of multiple applications
 - Improved performance for single threads
 - Area efficient implementation
 - Scalable performance and power
- Target implementations at 65nm and below
 - Dynamic and leakage power management



Mobile Processor Roadmap



Mobile Processor Roadmap



Falcon Processor

- Mid-range Cortex applications processor
 - 60% performance increase over ARM11
 - ARMv7 migration path from ARM11 processors
 - Jazelle RCT & DBX, Thumb-2, NEON, TrustZone and IEM technologies
 - Software compatible with Cortex-A8 processor
- Single core and MP versions
 - MPCore technology for cache-coherent multiprocessing
 - MP version will be AMBA 4 'ready' to support system coherency
 - Configurable option without NEON for targeted applications
 - Potential NEON integer only option for wireless baseband
- Product availability:
 - Lead partner release 3Q'07, General release 1Q'08

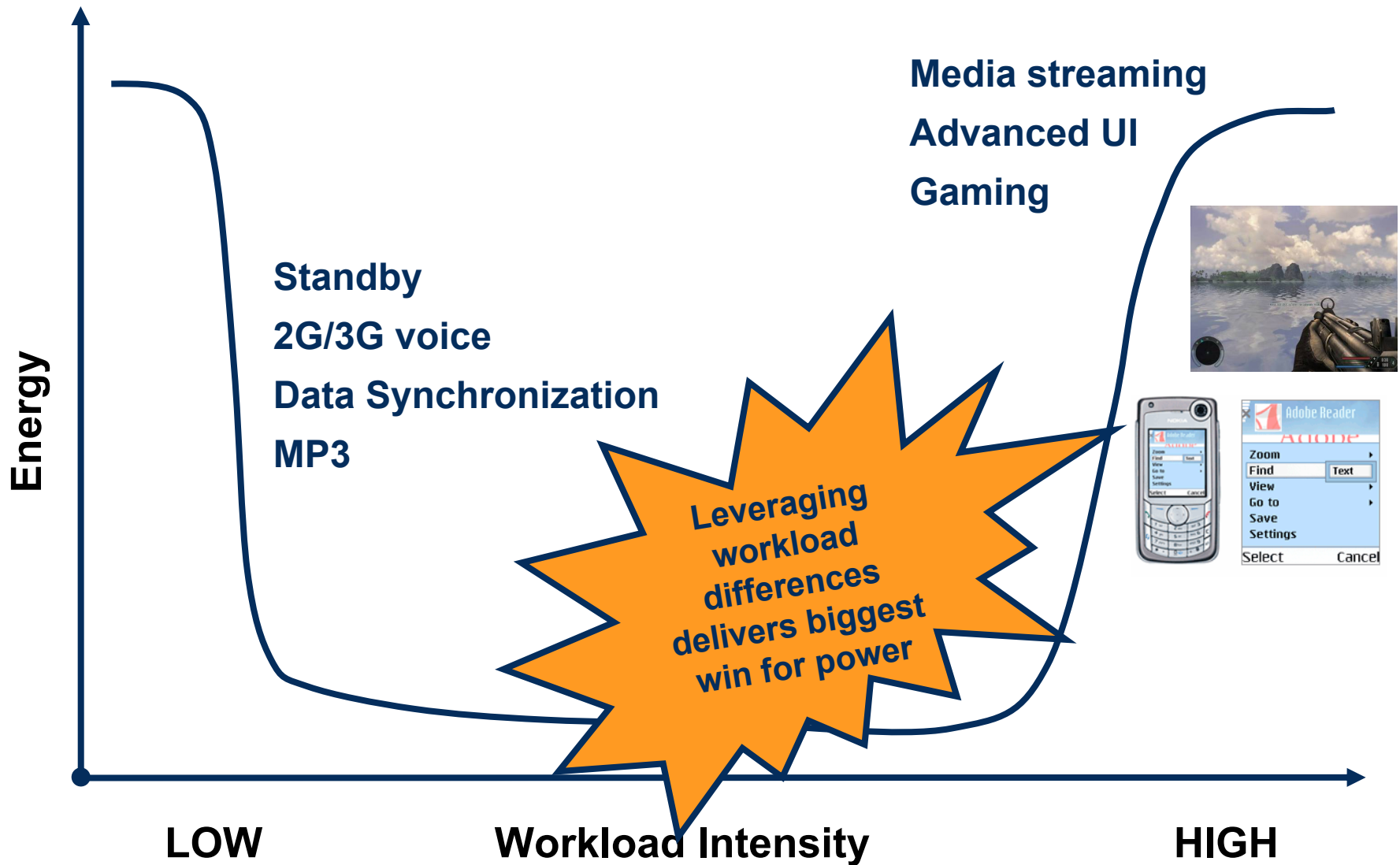


Eagle Processor Concept

- Eagle is based on Cortex-A8 with significant enhancements:
 - Coherency support
 - System coherency using AMBA 4
 - Single core and MP versions
 - Support for smaller companion core (Sparrow)
 - Sharing of the L2 Cache with other bus masters
 - Improvements in L1 and L2 memory system
 - Architectural enhancements (under consideration)
 - New security features
 - Need to explore virtualisation options
 - 16-bit float and Improved NEON FP
 - Support for optional configurations (NEON, FP)
- Performance improvements through efficiency
 - Designed to take full advantage of 45nm technology

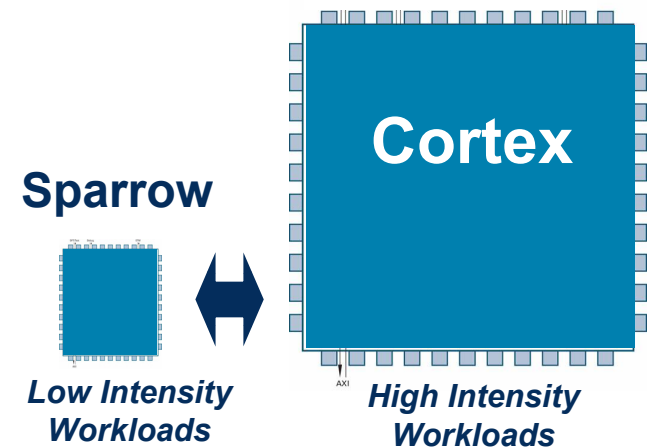


Wireless Workloads



Heterogeneous Multi-Processing

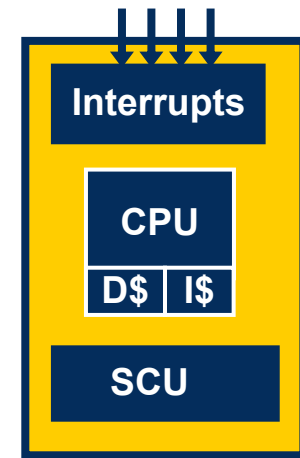
- Exploit wide variance in wireless workload
 - Processing transitions across higher performance processors and smaller companion core targeting 40% power savings
- Cores share common base architecture for OS
 - ARMv7, T2, TrustZone, Jazelle-RCT
- Processors support system-wide coherency
 - Larger core or multi-processor enables higher throughput
 - Smaller core sized to address low-power active processing
- Fast transition between cores
 - Additional engines for specialized processing



Heterogeneous Multi-Processing

- ARM is building system infrastructure to allow heterogeneous multi-processing with diverse processors and accelerators
 - The next frontier for low power after homogeneous multi-processing

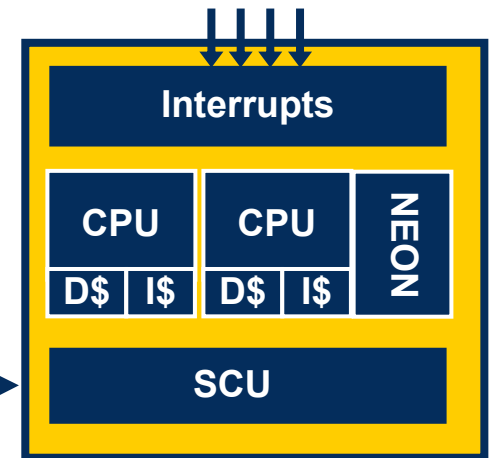
Low Intensity Workloads



Globally Coherent
Cached
Accelerators

Locally Coherent
Non-cached
Accelerators

High Intensity Workloads



Coherent AMBA 4 bus matrix

Shared System Cache

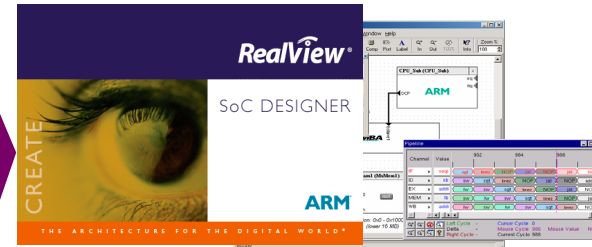
Bringing Your System to Market

- Today's systems are enormously complex
- Need to balance power, performance, area and time-to-market
- System level modeling enables early prototyping *and* platform optimization

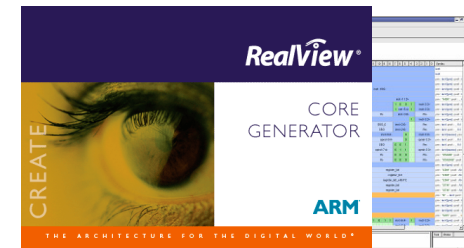
RealView®
Tools by ARM®

Then iterate
and optimize

Draft Initial System Design
Select ARM Components



Create New Components



Software Prototyping



Pre-Silicon SW and System Optimizations

Summary

- ARM technology is central to bringing next generation Handheld Multimedia Computing to consumers
- Enabling operators to provide flexible platforms and new services for consumers who appreciate the converged device
- Delivering high-performance and low-power systems requires not only leading edge CPUs but also fabric, tools, and physical IP
- At ARM, low-power is fundamental to all products and strategies
- ARM delivers best in class products from top-to-bottom

